Problem Set 10 - Files

Hello student,

the goal of this problem-set is to get started with files. Especially in your line of work, working with files is crucial.
Datasets are almost always presented as files. Therefore, it is important you need to learn to work with them.

Assignment 10.1 - BMI using files

Few weeks ago, in occasion of **assignment 8.2** you were asked to calculate a series of data concerning the BMI.
In that occasion you were provided a minimal set of data in the form of a list. However, this is rarely the case. Usually, this kind of data is provided using **files**.

The goal of this exercise is to do the same calculations, therefore re-using the code you already developed, but this time saving the results to files.
In particular, you will be asked to load the data from the provided `people.csv` file.

To load and save files you are only allowed to use Python's builtin modules and methods, i.e. no third part libraries such as Pandas, etc.

You should then produce as a result the following files:

- **bmi.csv** - containing all people with their BMI
- **highest_bmi.csv** - containing the person with the highest BMI
- **lowest_bmi.csv** - containing the person with the lowest BMI
- **average_bmi.csv** - containing the overall average, the males average and the females average

An example of the expected content of the files is reported here:

```
# bmi.csv
name,height,weight,gender,bmi
Anthony Harris,1.55,64.0,male,26.64
Donald Allen,1.56,76.0,male,31.23
Brian Green,1.65,66.0,male,24.24
Susan Mitchell,1.67,60.0,female,21.51
...

# highest_bmi.csv
name,height,weight,gender,bmi
Richard Garcia,1.5,109.0,male,48.44

# lowest_bmi.csv
name,height,weight,gender,bmi
Anthony Garcia,1.94,45.0,male,11.96

# average_bmi.csv
overall,males,females
26.49,26.47,26.52
```

Hint: https://docs.python.org/3/library/functions.html#open
Hint 2: https://docs.python.org/3.6/library/csv.html

Assignment 10.2 - Extract URLs

You are a data-scientist working at Microsoft. Your boss Ronald O'Brian asks you to analyze the main page of your website, because he wants to understand what links are contained.
The idea is to check every link that is still active. However, there are many, and the page source is pretty long. Therefore you decide to write a piece of code to extract and save them to a file for later analysis.

Your boss provides the function `extract_urls` (available in file `extract_urls.py`), which given a string, returns a list containing all the URLs or an empty list if no URLs were found.

Using the aforementioned function, write a software that does the following:

- Extracts URLs reading line by line the provided `microsoft.com.html` file;
- Prints the number of URLs found on the original file;
- Eliminates duplicates;
- Prints the number of URLs removed.
- Saves the resulting unique URLs to a file called `urls.txt` (one URL per line!).

An extract of the `urls.txt` you should obtain is the following:

```
https://www.microsoft.com/mscomhp/onerf/signout?pcexp=True
https://img-prod-cms-rt-microsoft-com.akamaized.net/cms/api/am/imageFileData/
    RE4E4rT?ver=2072&amp;q=90&amp;m=6&amp;h=195&amp;w=348&amp;b=%23FFFFFFFF&amp;
    l=f&amp;o=t&amp;aim=true
https://swiftkey.com/images/misc/stores/app/en.png
...
```

Please note that the file provided is encoded using UTF-8 (if you are working on windows you should probably pass the `encoding='UTF-8'` param to the `open()` function).

Assignment 10.3 - File merge

We want to build a function that is able to merge together the content of two text files.

The function must provide the following features:

- two parameters to pass the files we want to merge
- a parameter to specify the name of the output file (the one that will contain the merged result)
- an optional parameter to configure if we want to merge in an interleaved way (one line from the first file, one from the second, and so on) or not.
  When using the interleaved mode, make sure that if the first file has no more lines but the second does, the function works properly. In any case, all the content of the two files must be present in the resulting merged file.

Create then a main to test the correct behavior of your function, using the provided text files as inputs (`text_file_merge_file1.txt` and `text_file_merge_file2.txt`).

Assignment 10.4 - Read and sort file

Implement a program that is able to read and sort the data present in the provided file `data.txt`.

This file contains in random order, strings and numbers.

The program must first read the content of the file, storing it in two separate lists, one for each data type. After that, this data must be sorted in a descending way and wrote to two textual files.
At the end, the program must show the total number of elements read by the input file, the average of the numbers and a string formed by the concatenation on the first letter of each word.

Use the concepts discussed during the lecture to organize your code as best as possible (e.g. functions,...).

`Hint`: to find out if a line contains a number you could use the `isnumeric()` string method (works only for positive integer numbers).

Assignment 10.5 - Settings migration

You are a developer at Respawn Entertainment. You are working on a new version of the game engine. In this new version you are going to save user configurations using `pickle`, whereas before you were doing the same but with textual files.

Your job as a developer is to create a script for easy migration between the old textual format and the new pickle.

Write a program that does the following:

- Reads the old configuration file line-by-line;
- Cleanes each line of the useless char (i.e. `\n` and `"`);
- Creates a dict for each line where the keys are: `bind`, `key` and `action`;
- Saves the list of settings using `pickle`;
- Re-loads settings using `pickle`;
- Prints in a proper way the result.

You are provided an example of the old configuration file, in the form of the `settings.cfg` file, which contains the following:

```
bind_US_standard "1" "weaponSelectPrimary0"
bind_US_standard "2" "weaponSelectPrimary1"
bind_US_standard "4" "+scriptCommand4"
...
```

The list of dicts generated must look like this:

```
[
    {'bind': 'bind_US_standard', 'key': '1', 'action': 'weaponSelectPrimary0'},
    {'bind': 'bind_US_standard', 'key': '2', 'action': 'weaponSelectPrimary1'},
    {'bind': 'bind_US_standard', 'key': '4', 'action': '+scriptCommand4'}
    ...
]
```

And the settings, printed after loading the new pickle format, must look like:

```
bind: bind_US_standard key: 1 action: weaponSelectPrimary0
bind: bind_US_standard key: 2 action: weaponSelectPrimary1
bind: bind_US_standard key: 4 action: +scriptCommand4
...
```